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SUMMARY

Data scientist and full-stack engineer with 8+ years of professional experience across software engineering, security, and data science, now deploying production ML and analytics systems for utility operations. Current work includes architecting RPRS as sole engineer, building a Relay Failure Intelligence Dashboard, and contributing to an Equipment Prioritization Predictive Model. Earlier deployed an LLM-powered work order classification engine to production (March 2025) and delivered a call center forecasting system for Baltimore Gas and Electric through Exelon Academy. MS in Computer Information Science (Western Illinois), Cybersecurity minor; Security+ and CEH certified.

EXPERIENCE

EXELON CORP (ComEd)

July 2024 - Present

Data Scientist - Real Time Analyst

- Automated classification of thousands of Corrective Maintenance work orders by deploying an LLM-powered engine that predicts Equipment Type, Failure Mode, and Root Cause codes from free-text descriptions, then matches each record to the correct Relay Identification Number in the equipment master database. Engineered the end-to-end pipeline spanning enterprise work order system extraction, equipment master joins, LLM inference with confidence scoring, and SME validation. Co-designed the validation workflow with field engineers and reliability analysts. Achieved sub-second per-record processing feeding Transmission and Substation analytics.
- Improved root-cause data completeness 6x (10% to 60%), established 100% Relay Identification Number traceability where the prior process had no automated mapping, and reduced SME manual coding effort by over 90% after deploying the classification engine to production in March 2025. Validated against an SME-labeled ground truth dataset (93% Equipment Type, 80% Failure Mode, 73% Root Cause, 81.7% weighted accuracy). Reported results to leadership, with planned expansion across additional operating divisions and Exelon operating companies.
- Enabled daily staffing decisions at Baltimore Gas and Electric (BGE), an Exelon operating company, by designing and deploying a multi-model call center volume forecasting system that replaced a basic regression baseline. Engineered ~45 features spanning cyclic encodings, holiday adjacency, weather regressors with lags, and nine target-lag windows with rolling statistics. Trained a three-model ensemble (Prophet, XGBoost on log1p-transformed targets, Random Forest) validated through 5-fold TimeSeriesSplit cross-validation for 4-day forecasts. Processed ~14M historical call records into a live Power BI dashboard consumed daily by Care Center leadership.
- Architected a unified analytics database from scratch and built the data migration pipeline that extracts relay event, failure, SCADA, and operational data from Oracle and other department-owned source systems - writing bulk SQL queries, performing data cleaning and ETL transformations, and defining the target schema in coordination with upstream data owners across multiple operational divisions. The consolidated data layer will support production ML systems, real-time analytics, and downstream reporting.
- Consolidated 5+ fragmented operational tools into a unified enterprise analytics hub, architecting and building RPRS as sole engineer. Engineered 91 REST endpoints across 55+ backend services, six integrated modules, and a dual-schema PostgreSQL 16 data model on a Next.js 16/React 18/TypeScript/.NET 8 stack. Designed NERC CIP-007-6-aligned security controls with three-tier RBAC, PRC-004 audit compliance, and automated Oracle and Microsoft Dataverse ingestion (OAuth 2.0/MSAL), partnering with relay engineering, compliance leads, and executive stakeholders to align the platform with operational workflows and regulatory requirements.

EXELON CORP (ComEd)

October 2023 - July 2024

Associate Engineer

REACTS Department

- Built a GIS-based analytics application to process and analyze geospatial data for infrastructure and environmental projects. Incorporated advanced features such as spatial intersections, CRS validation, and multi-threaded geodatabase loading, reducing the Product Design Link process from 3 days to 10 minutes and enabling employees to complete a previously 5-month task in hours.
- Delivered the Jurisdictional Analysis and Risk Mitigation System (JARM) in partnership with planning and engineering teams, generating operational and capital expenditure savings of ~\$500K and improving execution efficiency by 30% through predictive analytics for risk identification and resource optimization.

AMI Department

- Designed an analytical framework for transformer asset management by applying clustering and Dynamic Time Warping (DTW) to large-scale AMI smart meter data, identifying overload patterns and at-risk transformers; presented findings to operations teams to inform targeted preventive maintenance strategies that enhanced grid reliability and extended equipment life.

DER Department

- Automated power system validation and correction by developing a suite of Python tools with the CYME API (cympy), batch-processing and validating hundreds of grid models against external benchmarks.
- Implemented advanced scripts to fix model issues (e.g., hidden sources, standardized spotloads) and perform automated load allocation with power factor adjustments, partnering with DER engineers and analysts to integrate tools into daily workflows and improve data integrity and readiness for DER hosting capacity studies.

EXELON CORP (ComEd)

June 2023 - August 2023

Data Analytics/Engineering - Internship

- Built and deployed an AI-powered automation system that reduced manual data entry from 40 hours/week to under 10 minutes, applying Regex, Tesseract OCR, Google Geocoding, and CNN models to extract, classify, and structure unformatted engineering data.
- Developed end-to-end digitization strategies combining mapping software (Azure GIS, Azure RI, GE Smallworld, PNI) with Python and Java automation scripts, partnering with planning and engineering teams to integrate line drawings into geospatial databases and improve accessibility for infrastructure planning.

ALAT BY WEMA BANK PLC

January 2021 - December 2021

Security Engineer - Internship

- Conducted security risk assessments on 50+ lab devices in coordination with IT teams and deployed detection tooling (Metasploit, Nmap, OpenSSL) to enhance incident response, achieving a 47% reduction in security incidents.
- Implemented and optimized firewalls and IDS/IPS systems, reducing security risks by 20% while improving system efficiency and uptime by 25-30%.
- Designed and tuned network analysis workflows using Wireshark, Snort, and Suricata for intrusion detection across lab infrastructure, strengthening real-time threat visibility and response accuracy.
- Investigated and resolved 100+ security incidents, consistently restoring services within 2 hours and cutting downtime by 35% to ensure business continuity.

Software Engineer

- Developed a custom anti-virus application that strengthened endpoint protection by 75% and safeguarded organizational digital assets against malware threats.
- Led the design of an NFC-based data capture and transformation system, enabling automated data collection with a 10% improvement in accuracy and faster downstream analytics.
- Designed and executed software testing frameworks across project delivery cycles to ensure cross-system compatibility and maintain release quality at scale.
- Applied Agile and iterative development practices to deliver two large-scale projects under budget (20% cost reduction, 60% operational efficiency gain), collaborating with client stakeholders on requirements and delivery.

Depot Administrative Analyst - Internship

- Built SQL data pipelines processing 1M+ record datasets (5% accuracy improvement, 20% faster processing) and forecasting models for pricing strategies, working with leadership and cross-functional business teams to drive 15-25% gains in gross margins across marketing, product development, and sales initiatives.

TECHNICAL SKILLS

- Machine Learning & AI: PyTorch, scikit-learn, XGBoost, Random Forest, Prophet | LLM prompt engineering, confidence scoring, SME validation loops | Pandas, NumPy, time-series forecasting (TimeSeriesSplit CV, recursive + direct-horizon), feature engineering, natural language processing
- Full-Stack Engineering: TypeScript, JavaScript, Python, C#, Java, SQL, R | Next.js, React, Tailwind CSS | .NET 8 Minimal API, REST API design, middleware architecture
- Data Infrastructure: PostgreSQL, Oracle, MySQL, MongoDB | EF Core 8, Apache Spark | Redis (L2 caching), IndexedDB | ETL pipelines, schema migrations, dual-schema architecture, OAuth 2.0/MSAL ingestion
- Cloud & DevOps: AWS (EC2, S3, Lambda, DynamoDB), Azure (GIS, AD, DevOps Git), Google Cloud (Vertex AI) | GitHub Actions CI/CD | Test frameworks: Playwright, Jest, xUnit, k6 | Git, Linux, macOS, Windows
- Security & Compliance: NERC CIP-007-6, PRC-004 workflows | JWT authentication, three-tier RBAC, CSRF protection, FluentValidation, rate limiting, immutable audit logging | Metasploit, Nmap, Wireshark, Snort, Suricata, OpenSSL | IDS/IPS, firewalls, network analysis
- Analytics & Reporting: Power BI (DAX), Plotly, Tableau | Geospatial analytics (Azure GIS, GE Smallworld, PNI, CYME API/cympy) | Jupyter Notebooks, Matplotlib

CERTIFICATIONS

- CompTIA Security+ | Certified Ethical Hacking from zSecurity | CCNA | Google Data Analytics | IBM - Supervised Machine Learning (including Regression) | IBM - Unsupervised Machine Learning | IBM - Exploratory Data Analysis for Machine Learning | Duke University - Managing Big Data with MySQL | Advanced Data Analysis with Generative AI | Machine Learning with Big Data | Spatial Analysis and Satellite Imagery in a GIS

RESEARCH PAPERS

- Comprehensive Analysis of SCADA System Data for Intrusion Detection Using Machine Learning (first author). GJETA, 2025.
- Explainable AI for Automated Threat Hunting in Large-Scale IoT Ecosystems (first author). IJCRT, 2025.
- Predictive Analytics for Aging U.S. Electrical Infrastructure Leveraging Machine Learning to Enhance Grid Resilience and Reliability (first author). WJARR, 2023.
- AI-driven advanced python pipeline for windows operating systems vulnerability detection and exploitation (co-author). MSARR, 2024.
- Post-Quantum Cryptographic Algorithms for Secure Communication in Decentralized Blockchain and Cloud Infrastructure (co-author). IJCATR, 2022.

LEADERSHIP & ACTIVITIES

- Invited Peer Reviewer, ICLR 2025 Workshop on XAI4Science (OpenReview) - reviewed submissions on explainable AI and scientific discovery
- Invited Peer Reviewer, Carnegie Mellon University LTI Student Research Symposium 2025 (OpenReview) - reviewed four NLP and ML submissions including LLM-based formal verification research
- Judge - eCYBERMISSION Army Educational Outreach Program; Globee Awards (AI & Cybersecurity)
- Reviewer - SWE Pathfinder Award, 2025 WE Local Prague Poster Competition, First Year Scholars Scholarship
- Member: NSBE, ACM, ISC2, ASCE, ISSA
- Participated in JOHN DEERE Cyber Tractor Challenge

EDUCATION

Master’s Degree in Computer and Information Science, minor in Cybersecurity

ALLOVER CENTRAL POLYTECHNIC